Clémence PREVOST

PhD Student

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RESEARCH INTERESTS

Multimodal data fusion, Inverse problems, Tensor decomposition, Low-rank approximation, Cramér-Rao bounds.

EDUCATION

• **PhD Student (Sept. 2018 —):** '*Multimodal data fusion by low-rank approximation*'. University of Lorraine, CRAN UMR 7039, CNRS, under the supervision of David Brie (Professor) and Konstantin Usevich (CNRS Research officer).
• **MSc (2017- 2018):** '*Complex Systems Engineering*'. University of Lorraine.
• **Engineering school (2015 - 2018):** '*Digital Systems Engineering*'. Ecole Nationale Supérieure d'Electricité et de Mécanique (ENSEM), University of Lorraine. Subjects studied; **Signal Processing**, **Applied Mathematics** and **Computer Science**.
• **CPGE (2013 - 2015):** PCSI - PSI*, Lycée César Baggio, Lille. Undergraduate studies to prepare for competitive entry exams to engineering schools. Subjects studied: **Mathematics**, **Physics**, French literature and English.
• **French Baccalaureate with honors (2013):** Lycée Denis Diderot, Langres. Subjects studied: **Biology**, **Physics**, **Mathematics** (speciality in **Linear algebra**).

COMMUNICATION IN CONFERENCES AND WORKSHOPS

International conferences

• '*Cramér-Rao Lower Bounds with random equality constraints*', Prévost, C., Chaumette, E., Usevich, K., Brie, D. and Comon, P. 2020 IEEE ICASSP (International Conference on Acoustics, Speech and Signal Processing).
• '*Cramér-Rao Bounds in the framework of hyperspectral super-resolution*', Prévost, C., Usevich, K., Haardt, M., Brie, D. and Comon, P. 2019 IEEE CAMSAP (international workshop on Computational Advances in Multi-sensor Adaptative Processing).
• '*Coupled tensor low-rank multilinear approximation for hyperspectral super-resolution*', Prévost, C., Usevich, K., Brie, D. and Comon, P. 2019 IEEE ICASSP (International Conference on Acoustics, Speech and Signal Processing).

National conferences

- '*Approches tensorielles couplées pour la fusion aveugle d'images multispectrale et hyperspectrale*', Prévost, C., Usevich, K., Brie, D. and Comon, P. 2019 GRETSI (Colloque francophone de traitement de signal et des images).

Workshops and seminars

- '*Hyperspectral super-resolution via coupled Tucker decomposition*', Prévost, C., Usevich, K., Brie, D. and Comon, P. 2019 GdR ISIS meeting on tensor decompositions.
- '*Hyperspectral super-resolution via coupled Tucker decomposition*', Prévost, C., Usevich, K., Brie, D. and Comon, P. 2019 LMRA (Workshop on Low-Rank Models and Applications):
- '*Coupled tensor low-rank multilinear approximation for hyperspectral super-resolution*', Prévost, C., Usevich, K., Brie, D. and Comon, P. 2019 Peyresq Summer School.

PUBLICATIONS

- **'***Hyperspectral super-resolution with coupled Tucker approximation: Identifiability and SVD-based algorithms***',** IEEE Transactions on Signal Processing, vol. 68, p.931-946. Prévost, C., Usevich, K., Brie, D. and Comon, P.

TEACHING

- **2018 - 2019:** Practical courses: study of first and second order filters, introduction to the FFT and frequency analysis, frequency modulation (64h). IUT Nancy-Brabois, Department of Telecom and Network (R&T).
- **2019 - 2020:** Practical courses: study of first and second order filters, introduction to the FFT and frequency analysis (64h). IUT Nancy-Brabois, Department of Telecom and Network (R&T).

PROJECTS

- *'Coupled Tensor Decomposition for the Fusion of Variable Multimodal Images: Identifiability and Hyperspectral Super Resolution'*, funded by **GdR ISIS** mobility grant. This project involves researchers and student from CRAN, Nancy (David Brie, Konstantin Usevich, Clémence Prévost) and Laboratoire Lagrange, Nice (Cédric Richard, Ricardo Borsoi). The aim is to develop tensor-based methods for solving the hyperspectral super-resolution problem accounting for spectral and spatial variability between a hyperspectral and a multispectral image of the same scene.

ORGANIZATION OF CONFERENCES AND WORKSHOPS

- **ENBIS-18, Nancy, France:** Student organizer - Preparation, registration desk, technical chair (~8h).
- **IEEE CAMSAP-19, Le Gosier, Guadeloupe:** Student organizer - registration desk (4h).